



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to KURINJIRAJA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

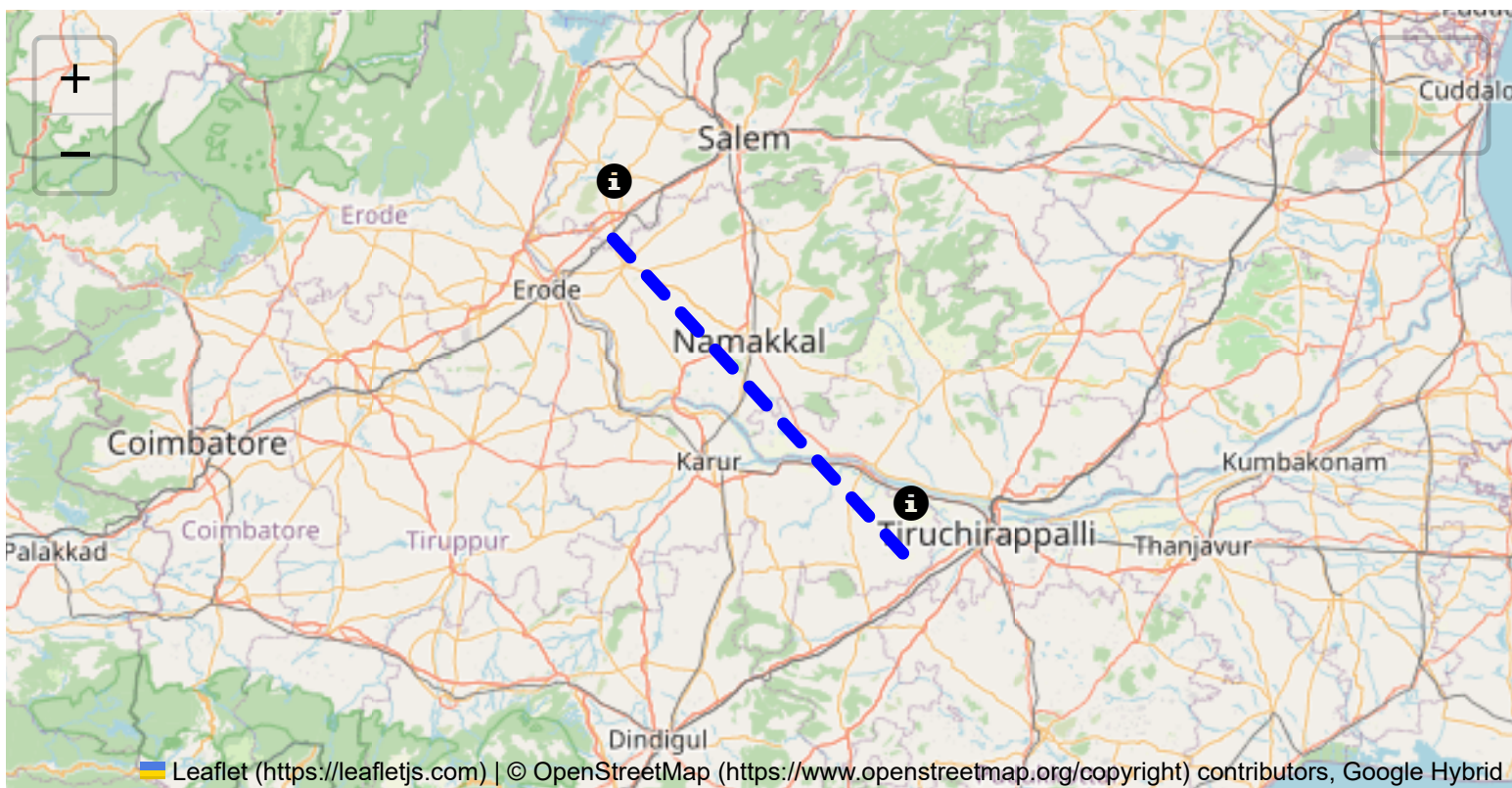
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

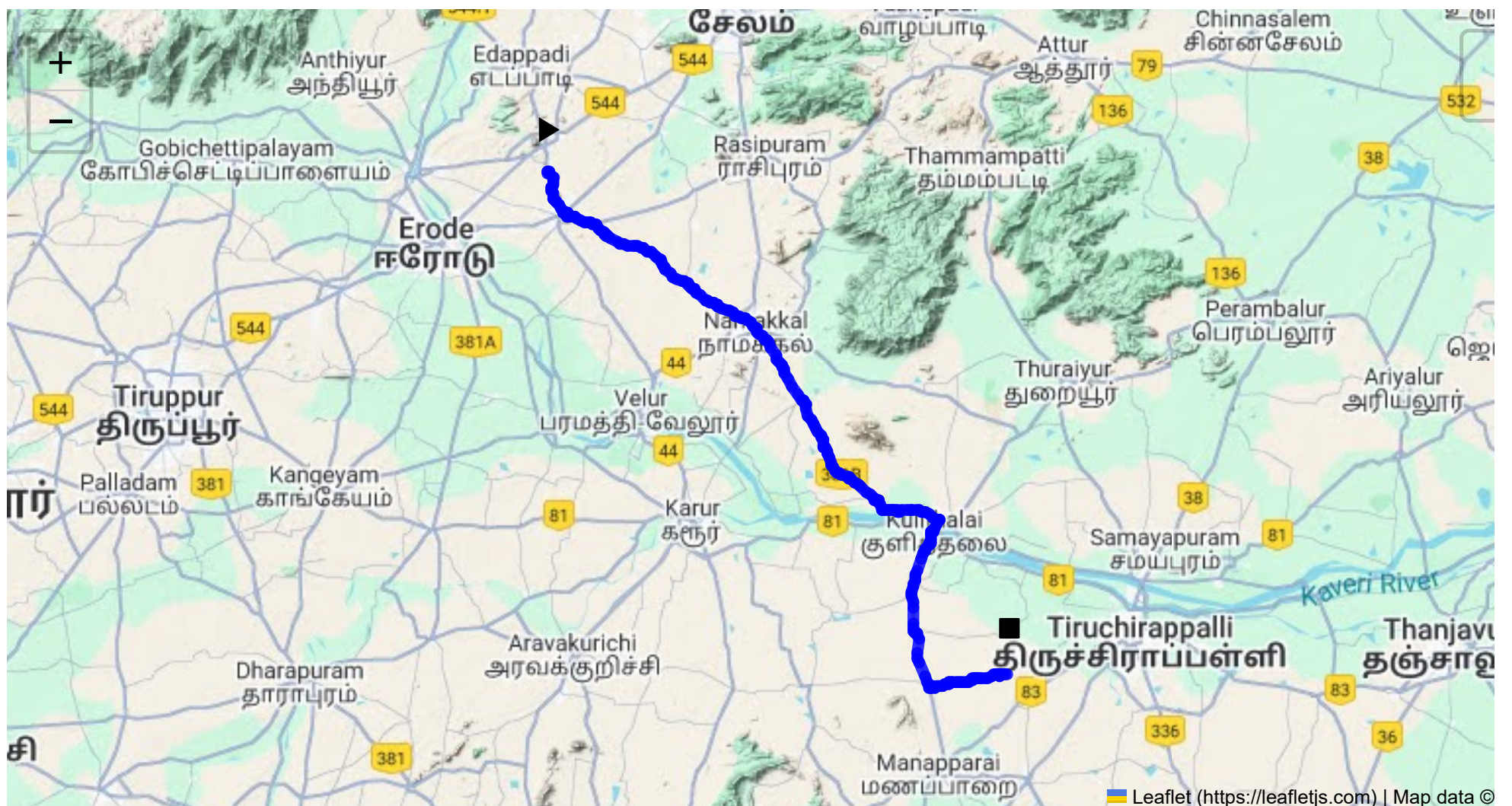
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 129.12 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.4 hours
Start: (11.4381, 77.8734)
End: (10.7484542, 78.51972)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri (CVQF+23W) to Kavalkaranpatti (PGX9+9V) spans approximately 129.12 kilometers. It likely involves major highways such as NH 44 and various state highways, traversing through semi-urban and rural areas, including towns and agricultural landscapes. Preliminary research suggests several small towns that might be passed, including Edappadi and Mallasamudram.

Weather Conditions and Hazards

Tamil Nadu experiences a tropical climate with distinct wet and dry seasons. During the southwest monsoon (June to September) and northeast monsoon (October to December), heavy rains might cause slippery roads and flooding in low-lying areas. Summer months (March to June) can be very hot, which might affect vehicle performance, especially for trucks.

Traffic Patterns

Traffic tends to be heavier during morning (8 AM - 10 AM) and evening (5 PM - 8 PM) rush hours, particularly in larger towns and near industrial areas. The NH 44 is a major thoroughfare and might experience congestion. Increased vigilance is needed at irregular road junctions and town areas where local traffic may be dense.

Road Quality and Infrastructure

The road quality varies along the route. Highways like NH 44 are generally well-maintained, but smaller state highways or rural roads may have uneven surfaces, potholes, and limited shoulder space. Watch for narrow bridges and underpasses.

Alternative Routes for Emergencies

In emergencies, consider taking parallel state highways, though these can be narrower and might not be suitable for heavy vehicles. Local roads can serve as temporary detours, but verify their condition and capacity for carrying heavy loads in advance.

Local Regulations on Hazardous Material Transport

Tamil Nadu has regulations governing the transport of hazardous materials that require specific permits and adherence to safety protocols. Ensure all necessary documentation is obtained, and the vehicle is labeled appropriately in accordance with the Motor Vehicles Act and the Central Motor Vehicles Rules.

Historical Incidents

While there are no major specific reports easily accessible about historical incidents involving hazardous materials along this exact route, highways such as NH 44 have seen incidents involving heavy vehicles. This necessitates careful adherence to traffic rules and precautions, especially in high-density traffic areas.

Environmental Considerations

The route crosses several rivers and streams; hence, spills could have serious environmental impacts. Agricultural areas are also present, and trucks should avoid unnecessary idling near fields to prevent crop damage due to exhaust emissions.

Communication Coverage

Generally, mobile networks cover most of the route. However, rural areas may experience occasional dead zones. Carry a satellite phone or ensure drivers know where reception is limited.

Estimated Emergency Response Times

Emergency services can take anywhere from 15 minutes to over an hour to reach certain rural or remote stretches, depending on proximity to a town or city. It's crucial to know the locations of nearby hospitals and fire stations along the route.

Overall Risk Assessment

The route entails moderate risks typical of highway transport in semi-urban and rural India. Major concerns are related to variable road conditions, weather impacts, and traffic congestion in towns. Drivers should be prepared for emergency rerouting and maintain strict adherence to safety standards and regulatory requirements. Regular communication and real-time updates on road situations will further enhance safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.94342, 78.41370	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44019, 77.87547	15 KM/Hr
4	Turn	Medium	11.39685, 77.88861	30 KM/Hr
5	Turn	Medium	11.38376, 77.89914	30 KM/Hr
6	Turn	High	11.38186, 77.90258	15 KM/Hr
7	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
8	Turn	Medium	11.33425, 78.00843	30 KM/Hr
9	Turn	Medium	11.22411, 78.16637	30 KM/Hr
10	Turn	Medium	11.21992, 78.16759	30 KM/Hr
11	Turn	Medium	11.12316, 78.23249	30 KM/Hr
12	Turn	Medium	11.12001, 78.23557	30 KM/Hr
13	Turn	Medium	10.98686, 78.33767	30 KM/Hr
14	Turn	Medium	10.97256, 78.34408	30 KM/Hr
15	Turn	High	10.96031, 78.42222	15 KM/Hr
16	Turn	High	10.96084, 78.42307	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
17	Blind Spot	Blind Spot	10.96050, 78.42371	10 KM/Hr
18	Turn	Medium	10.80330, 78.39098	30 KM/Hr
19	Turn	Medium	10.80307, 78.39192	30 KM/Hr
20	Turn	High	10.72760, 78.40999	15 KM/Hr
21	Blind Spot	Blind Spot	10.72725, 78.41135	10 KM/Hr
22	Turn	Medium	10.73076, 78.41216	30 KM/Hr
23	Turn	Medium	10.73163, 78.41346	30 KM/Hr
24	Turn	High	10.73156, 78.41546	15 KM/Hr
25	Turn	High	10.72931, 78.41759	15 KM/Hr
26	Turn	High	10.73300, 78.43553	15 KM/Hr
27	Turn	High	10.73519, 78.43623	15 KM/Hr
28	Turn	High	10.73555, 78.44066	15 KM/Hr
29	Turn	Medium	10.73660, 78.44129	30 KM/Hr
30	Turn	Medium	10.73725, 78.45414	30 KM/Hr
31	Turn	Medium	10.73780, 78.46697	30 KM/Hr
32	Turn	High	10.73753, 78.46866	15 KM/Hr
33	Turn	High	10.74164, 78.47182	15 KM/Hr
34	Turn	Medium	10.74026, 78.47579	30 KM/Hr
35	Turn	Medium	10.74180, 78.47873	30 KM/Hr
36	Turn	Medium	10.74545, 78.50928	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
6	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
15	hospital	Aruthra Hospital	11.2264666, 78.1631524	30 km/h	Medium
16	clinic	Sri Ram Nursing Home, Namakkal	11.2259986, 78.1631944	30 km/h	Medium
17	clinic	Thangavel Skin Care	11.2253765, 78.1644957	30 km/h	Medium
18	hospital	Rudhra Hospital	11.226569, 78.163216	30 km/h	Medium
19	hospital	Rajendran Children's Hospital	11.2251887, 78.1649142	30 km/h	Medium
20	clinic	Lung Care Centre	11.2276789, 78.1638586	30 km/h	Medium
21	hospital	Vignesh Hospital	11.224818, 78.1621315	30 km/h	Medium
22	hospital	Venkat Hospital	11.2253645, 78.1646159	30 km/h	Medium
23	hospital	Shree Meenakshi Dental Hospital	11.225269, 78.164275	30 km/h	Medium
24	hospital	Ilamurugan Hospital	11.2248399, 78.166586	30 km/h	Medium
25	hospital	Annai Eye Hospital	11.2241645, 78.1641366	30 km/h	Medium
29	clinic	Kumaran Poly Clinic	11.2233138, 78.1639212	30 km/h	Medium
30	clinic	Chandra Test Tube Baby Centre	11.2241806, 78.1645931	30 km/h	Medium
31	hospital	Aburvaa Ayurveda Hospital	11.2228807, 78.1640453	30 km/h	Medium
32	clinic	Vishnu Ayurveda Clinic	11.224262, 78.164173	30 km/h	Medium
33	clinic	Dhanvanthiri Physiotheraphy	11.2245256, 78.1638528	30 km/h	Medium
34	hospital	Aravinth Hospital	11.2233336, 78.1672805	30 km/h	Medium
35	hospital	Sri Nidhi Hospital, Namakkal	11.2255711, 78.1687291	30 km/h	Medium
36	clinic	Athithya Nursing Home	11.2237726, 78.1686447	30 km/h	Medium
37	hospital	Kathiravan Hospital	11.226651, 78.167351	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
38	hospital	Sri Ramana Hospital	11.223503, 78.1685433	30 km/h	Medium
39	hospital	Vanitha Child Care & Skin Hospital	11.2243378, 78.16811	30 km/h	Medium
40	hospital	Sri Dhanvanthries Hospital	11.2241181, 78.1678614	30 km/h	Medium
41	hospital	Nirmala Hospital	11.2179361, 78.1665001	30 km/h	Medium
42	hospital	Surya Hospital, Namakkal	11.2178506, 78.1660375	30 km/h	Medium
43	hospital	Shanthi Hospitals, Namakkal	11.2190661, 78.1685735	30 km/h	Medium
44	hospital	Ganga Hospital	11.21742, 78.170272	30 km/h	Medium
45	hospital	Government Hospital	11.2168924, 78.169068	30 km/h	Medium
47	police	Police Station	11.2185353, 78.1695347	30 km/h	Medium
48	police	Women Police Station	11.2183133, 78.1692677	30 km/h	Medium
49	hospital	Manoharan Hospital	11.218016, 78.171629	30 km/h	Medium
50	hospital	Saravana Hospital	11.216652, 78.171298	30 km/h	Medium
52	hospital	Latha Eye Hospital	11.2185689, 78.1719035	30 km/h	Medium
53	hospital	Vasan Eye Care Hospital, Namakkal	11.216301, 78.1717694	30 km/h	Medium
54	hospital	Sri Ragavendra Hospital	11.2151734, 78.1703431	30 km/h	Medium
55	hospital	Aishwarya Hospital	11.2164042, 78.1715972	30 km/h	Medium
56	hospital	Thangam Hospital,Namakkal	11.21472, 78.171857	30 km/h	Medium
57	hospital	Poornima Hospital	11.215533, 78.16911	30 km/h	Medium
58	hospital	Nagarajan Hospital	11.2171673, 78.1718618	30 km/h	Medium
59	clinic	Ramesh ENT & Skin Care	11.217133, 78.171071	30 km/h	Medium
60	hospital	SMK Hospital	11.2172759, 78.1706866	30 km/h	Medium
61	clinic	ENT Clinic	11.215786, 78.170545	30 km/h	Medium
62	clinic	Hari Chest Clinic	11.2162053, 78.1710093	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
63	hospital	Dr. Kandasamy Hospital	11.216051, 78.172052	30 km/h	Medium
64	clinic	Primecare Diabetic Clinic	11.213992, 78.1737091	30 km/h	Medium
65	clinic	Dr. Shanmugam ENT Clinic	11.218762, 78.171206	30 km/h	Medium
66	clinic	Children's Clinic	11.217004, 78.171504	30 km/h	Medium
67	clinic	Krishiv's Dental Care & Orthodontic Centre	11.2184886, 78.1713346	30 km/h	Medium
71	hospital	Dr. A. Nagarajan Children's Hospital	11.212765, 78.1747068	30 km/h	Medium
72	hospital	M.M. Hospital	11.2103089, 78.1776545	30 km/h	Medium
73	clinic	Ram Clinic	11.2090922, 78.1790782	30 km/h	Medium
74	clinic	Prasith Clinic	11.208902, 78.180957	30 km/h	Medium
75	hospital	Imayam Speciality Hospital	11.2078899, 78.1797861	30 km/h	Medium
76	clinic	Aburvaa Multi Special Herbal Clinic	11.207568, 78.183483	30 km/h	Medium
77	hospital	Sugam Children's Hospital	11.2060083, 78.1854033	30 km/h	Medium
79	hospital	Government Hospital, Valayapatty	11.1212591, 78.2315962	30 km/h	Medium
82	hospital	Saravana Hospitals	10.9944313, 78.3273838	30 km/h	Medium
83	hospital	Dr. Siva Hospital	10.9877652, 78.3359228	30 km/h	Medium
84	hospital	Dr. Alva Hospital	10.987225, 78.336154	30 km/h	Medium
85	hospital	Ravikumar Hospital	10.988317, 78.336248	30 km/h	Medium
86	police	Thottiyam Police station	10.9874415, 78.3363001	30 km/h	Medium
87	hospital	Government Hospital, Thottiyam	10.98388, 78.338648	30 km/h	Medium
88	hospital	Government Hospital	10.9428055, 78.4146362	30 km/h	Medium
89	hospital	T M R Hospital	10.9429825, 78.4158242	30 km/h	Medium
90	hospital	Thogamalai, Government Hospital	10.7268532, 78.4097304	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
7	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
8	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
12	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
26	school	Government Girls School	11.2238954, 78.1662583	30 km/h	Medium
27	school	Government Elementary School	11.224555, 78.163967	30 km/h	Medium
28	school	Government Secondary School	11.2244426, 78.1647533	30 km/h	Medium
46	school	Government Girls Higher Secondary School	11.2179186, 78.1672383	30 km/h	Medium
51	college	Veterinary College and Research Institute Hospital	11.2144944, 78.1722653	30 km/h	Medium
68	college	Namakkal Kavignar Ramalingam Government Arts College for Women	11.2125137, 78.1724654	30 km/h	Medium
69	school	Kandasamy Kandar Matriculation and Higher Secondary School	11.2137988, 78.1698234	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
70	school	Selvam Matriculation Higher Secondary School	11.2124286, 78.1770824	30 km/h	Medium
78	college	King college	11.1692622, 78.2037212	30 km/h	Medium
80	school	Government Higher Secondary School	11.0234395, 78.2859835	30 km/h	Medium
81	college	Vetri vinayaga institution	11.013503, 78.3072053	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 10.94342, 78.41370



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38376, 77.89914



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

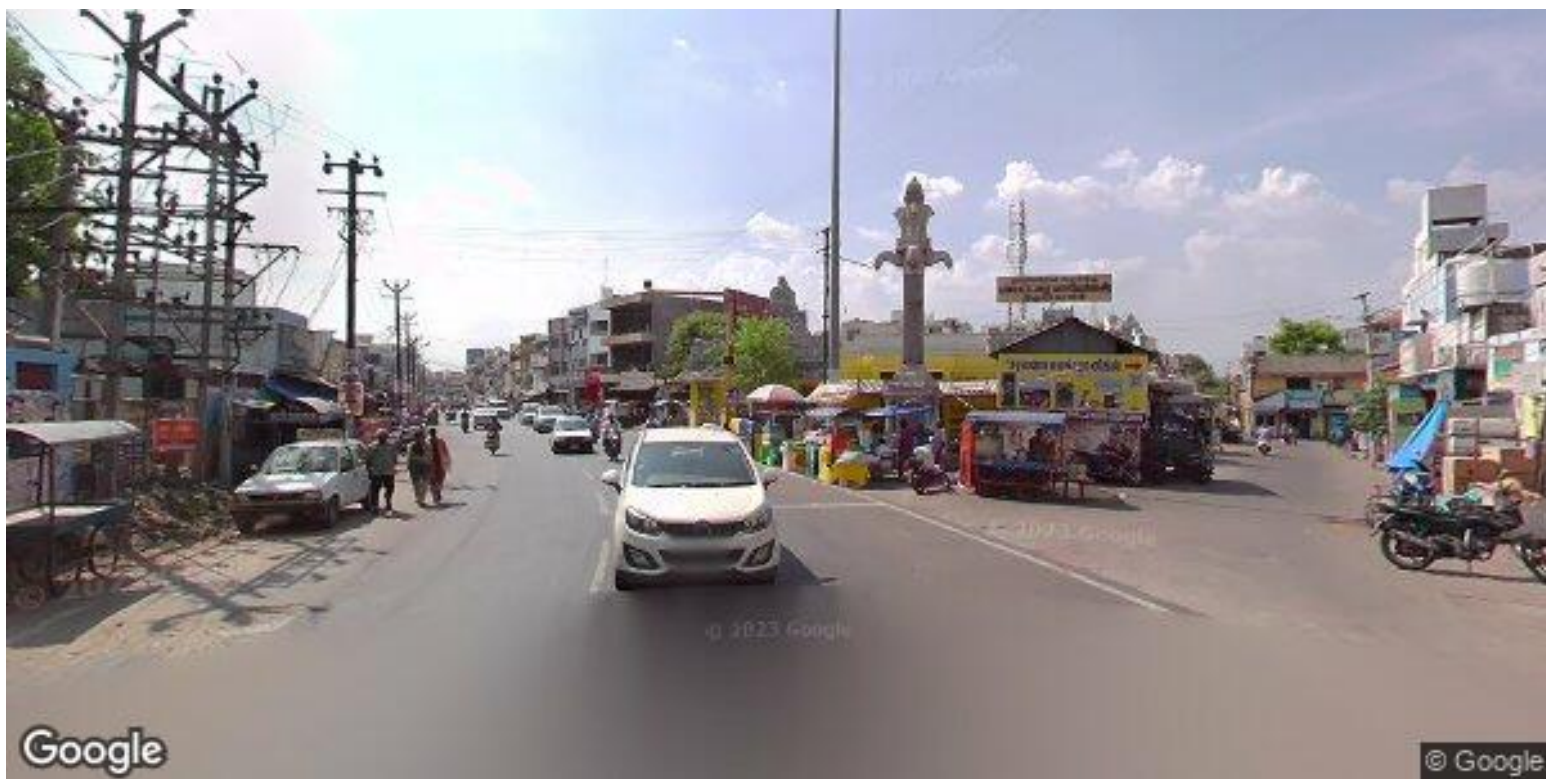
Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.33425, 78.00843



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.22411, 78.16637



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.21992, 78.16759



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.12316, 78.23249



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.12001, 78.23557



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.98686, 78.33767



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.97256, 78.34408



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.96031, 78.42222



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.96084, 78.42307



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.96050, 78.42371



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.80330, 78.39098



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.80307, 78.39192



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.72760, 78.40999



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 10.72725, 78.41135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.73076, 78.41216



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.73163, 78.41346



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.73156, 78.41546



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.72931, 78.41759



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.73300, 78.43553



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.73519, 78.43623



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.73555, 78.44066



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.73660, 78.44129



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.73725, 78.45414



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.73780, 78.46697



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.73753, 78.46866



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.74164, 78.47182



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.74026, 78.47579



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.74180, 78.47873



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.74545, 78.50928
